

Generation Interconnection System Impact Study Report

For Queue Project Q19-1270

Bpa Troutdale 100 kV

January 24, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1270. The Interconnection Customer proposes a 233.7 MW solar photovoltaic facility interconnecting at the Bpa Troutdale 100 kV substation in the Illinois area. The request was received on August 20, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 415 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,350,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LISLE - BPA TROUTDALE 100 kV	BPA TROUTDALE - SARIVAL 100 kV L	203.9	103	100.0
BPA TROUTDALE - SARIVAL 100 kV	LISLE - BPA TROUTDALE 100 kV	148.8	141	100.0
LISLE - TROPIC ISLE 100 kV	SARIVAL - BPA TROUTDALE 100 kV	118.1	103	61.2
SARIVAL - BRENAS 9201 100 kV	LISLE - BPA TROUTDALE 100 kV line	108.1	117	35.8
DOOVILLE - THOMAS HILL STATION 230 kV	MONTPELIER ELECTRIC GENERATING STA 500 kV/230 kV Auto	101.9	925	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LISLE - BPA TROUTDALE 100 kV	Rebuild line with higher-capacity conductor	\$2,104,000	100% (\$2,104,000)
BPA TROUTDALE - SARIVAL 100 kV	Rebuild line with higher-capacity conductor	\$1,010,000	100% (\$1,010,000)
TROPIC ISLE SUBSTATION - LISLE SUBSTATION 100 kV	Rebuild line with higher-capacity conductor	\$1,336,000	100% (\$1,336,000)
SARIVAL - BRENAS 9201 100 kV	Reconductor line	\$745,000	100% (\$745,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DOOVILLE - THOMA HILL STA 230 kV Line	Reconductor line	\$12,884,000	1% (\$155,000)
	Total allocated to project		\$5,350,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.